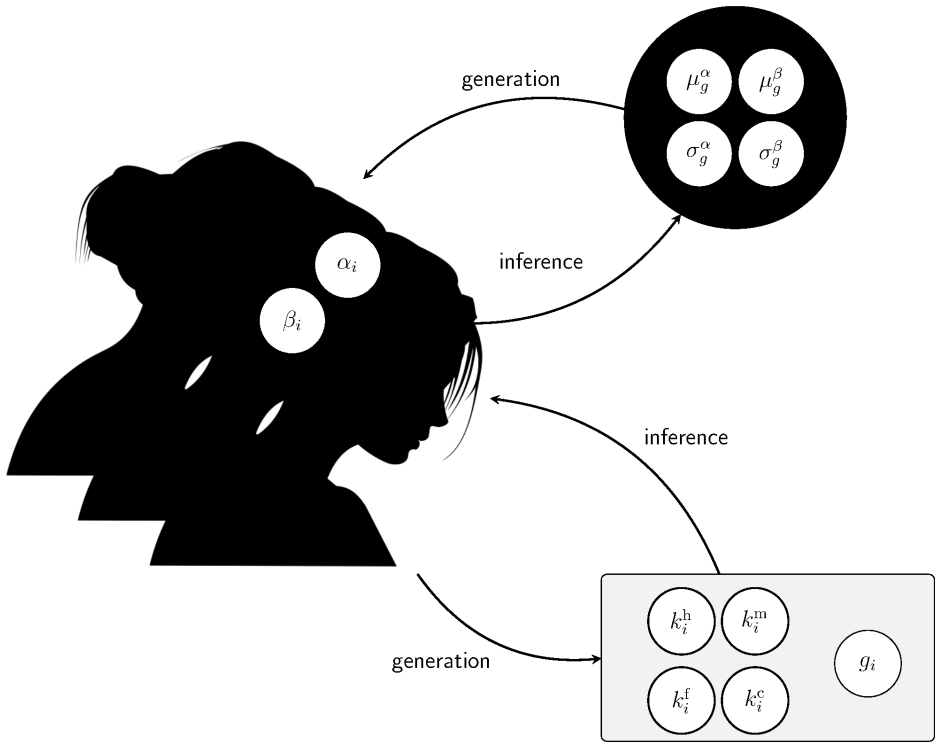
Hierarchical High Threshold Models

Michael D. Lee

Hierarchical High Threshold Model Structure

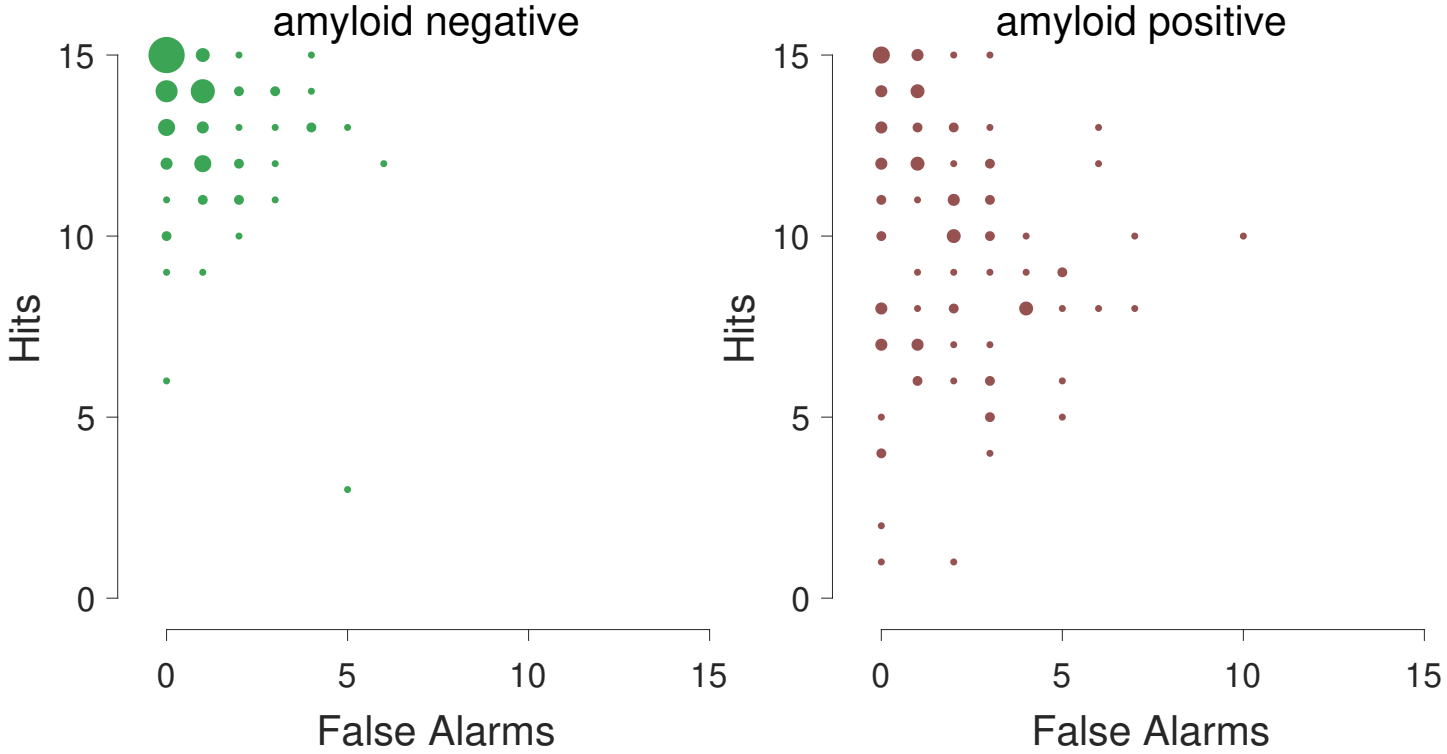
- The hierarchical high threshold models have the structure
  - observed counts of hits k_i^{h}, false alarms k_i^{f}, misses k_i^{m}, and correct rejections k_i^{c}
  - known allocation of each individual to a group g_i
  - individual detection α_i and guessing bias β_i parameters
  - the detection and guessing probabilities come from Gaussian group distributions with means μ_g^{α}, μ_g^{β} and standard deviations σ_g^{α}, σ_g^{β}



Amyloid Memory

Recognition Memory and Amyloid

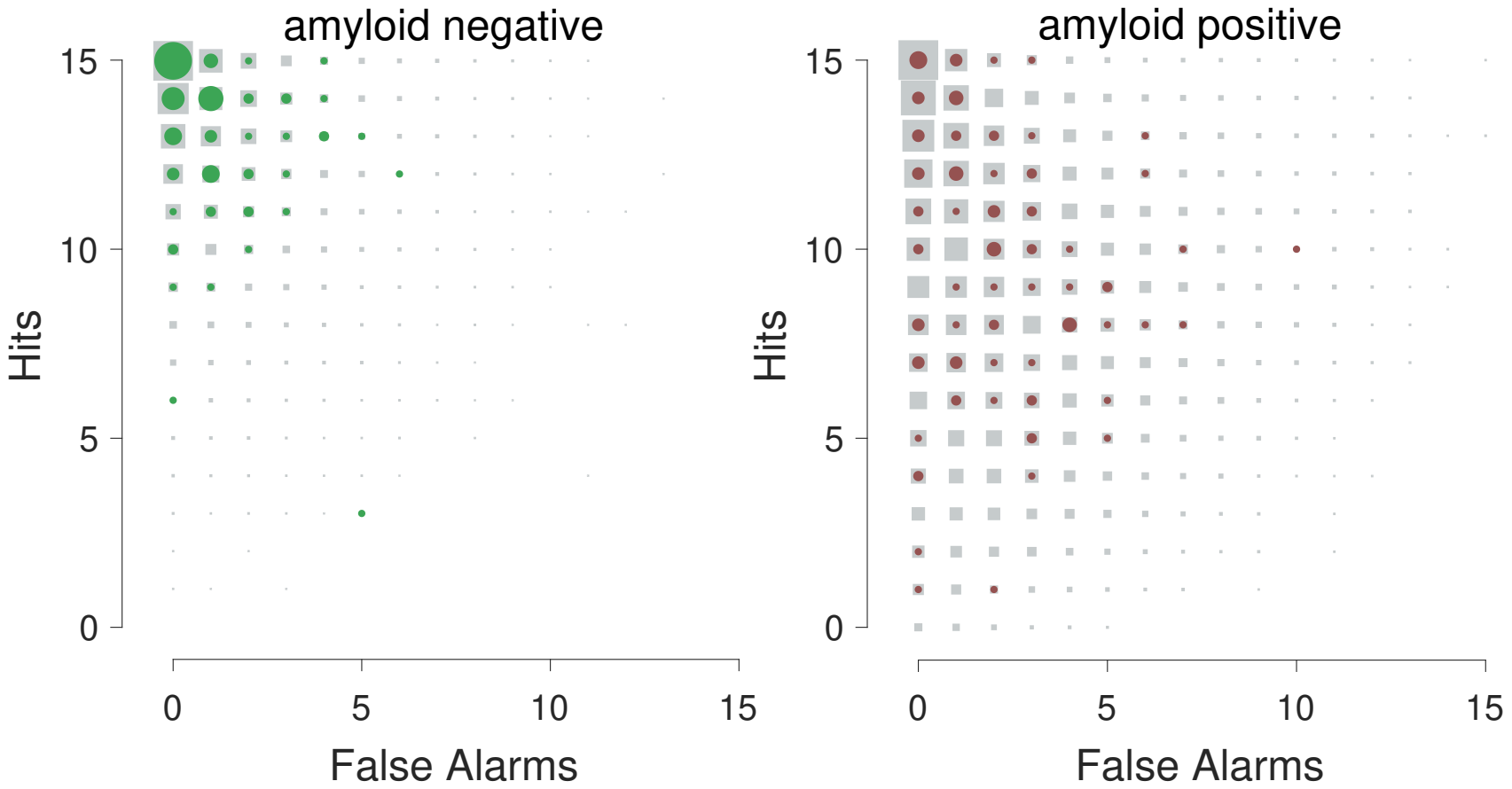
- 200 patients aged over 75 years were clinically assessed
  - memory: via the RALVT recognition memory task (Bean, 2011), with 15 old words and 15 new words
  - amyloid: classified as amyloid positive or negative, based on a cerebrospinal fluid measurement
- The hit and false alarm counts for each group are shown



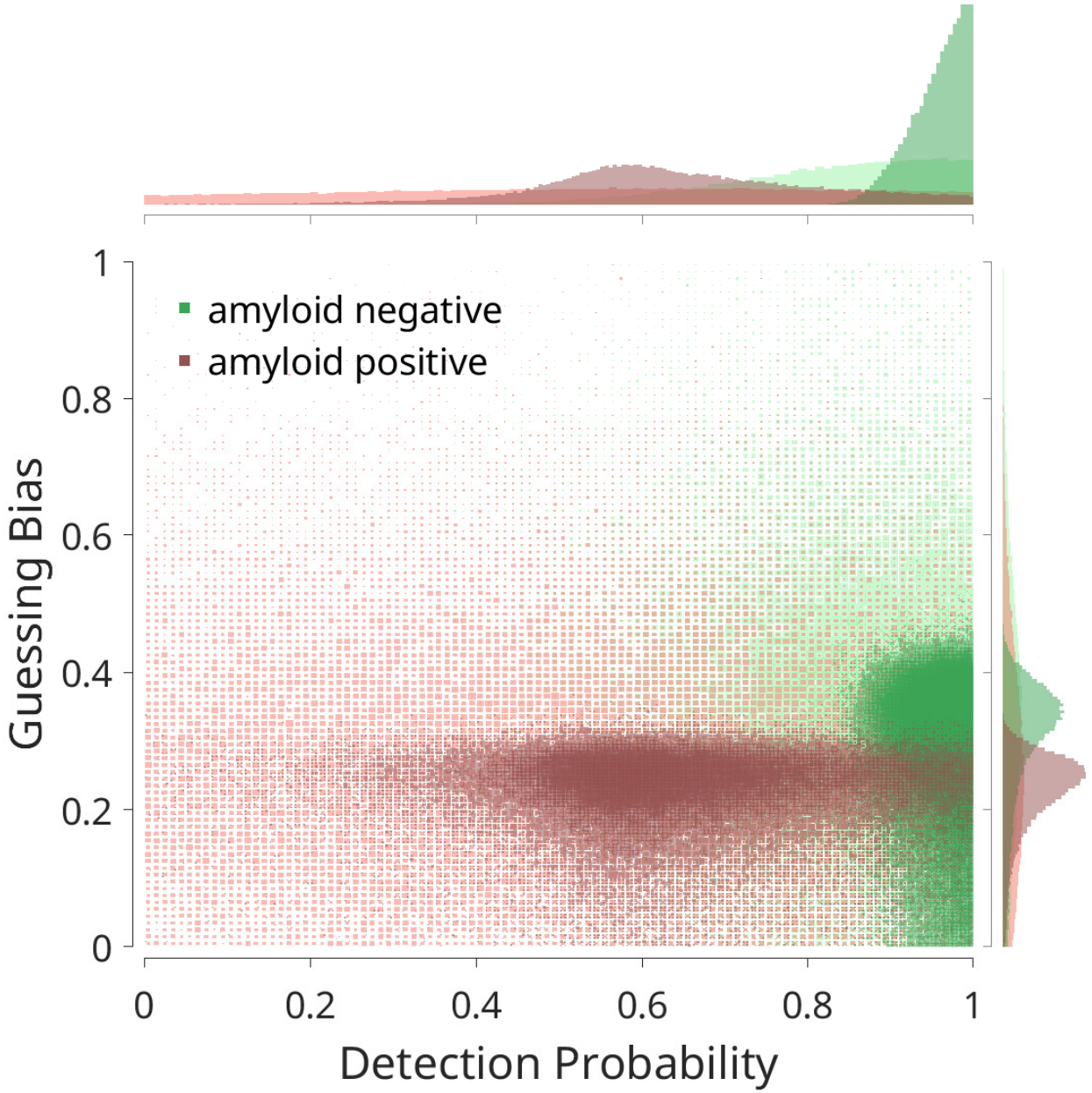
- We use the hierarchical two high threshold model

Descriptive Adequacy

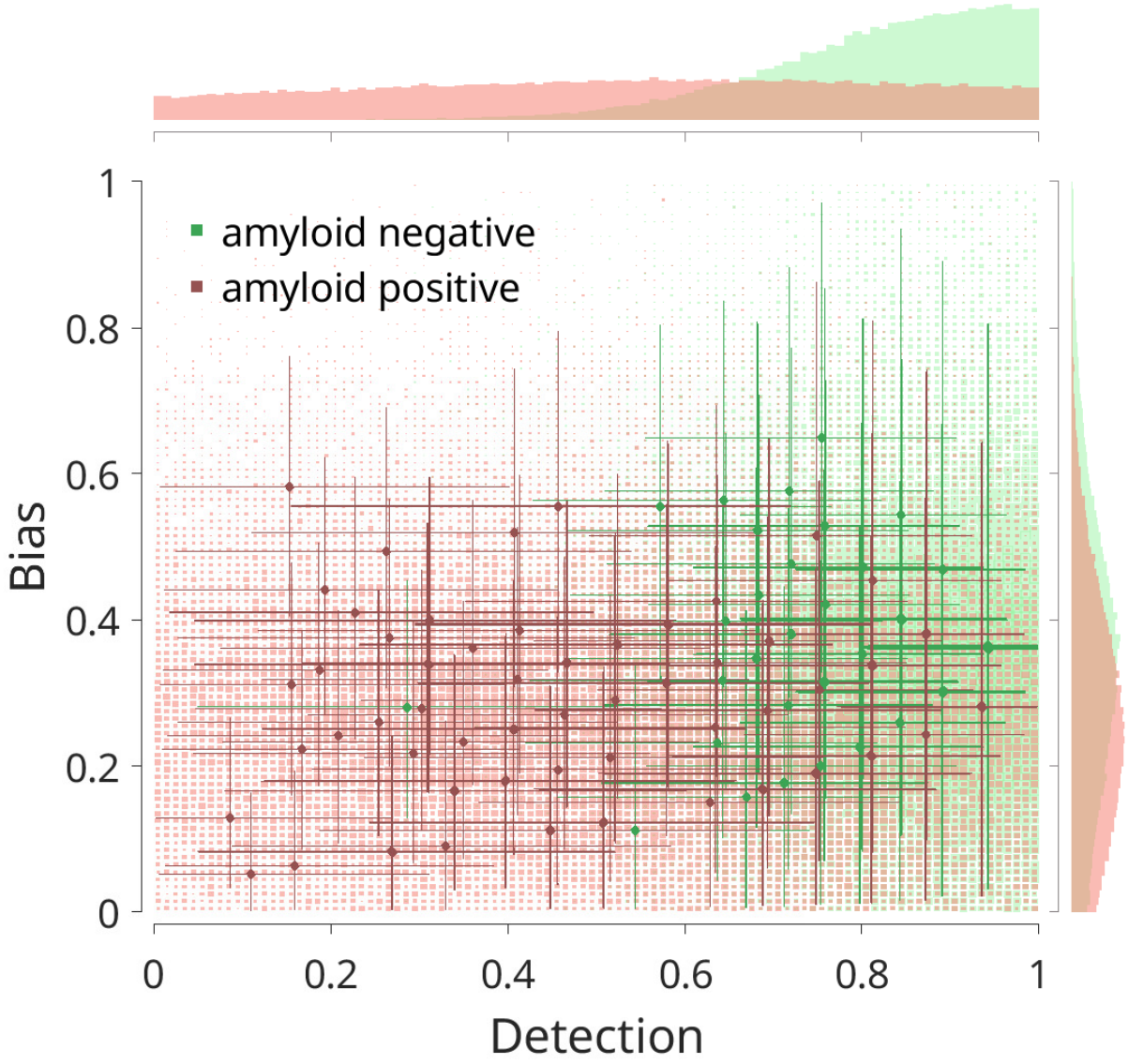
- The squares represent the posterior predictive distribution for the representative participant, and are overlaid on the individual data



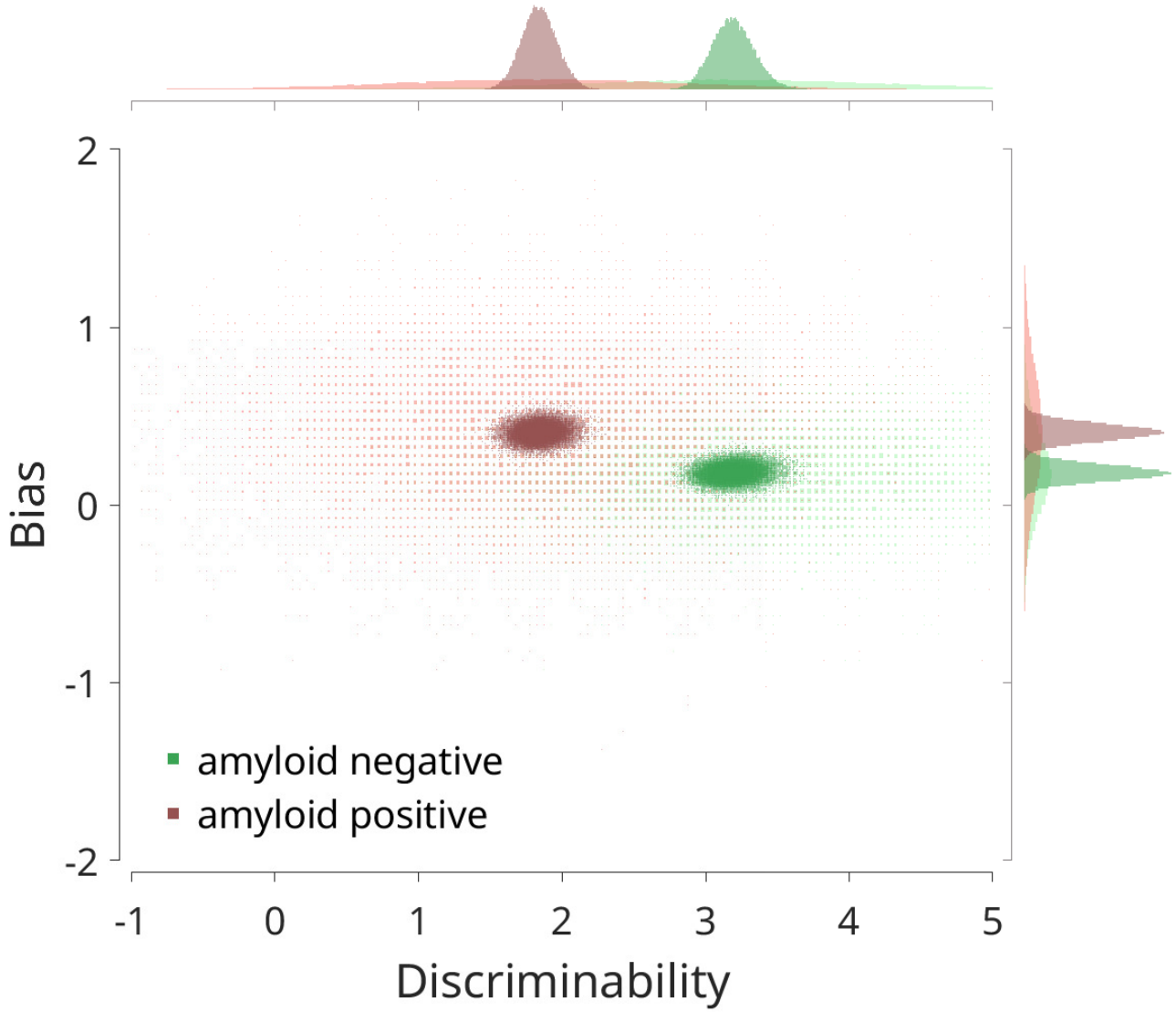
Group-level Inferences



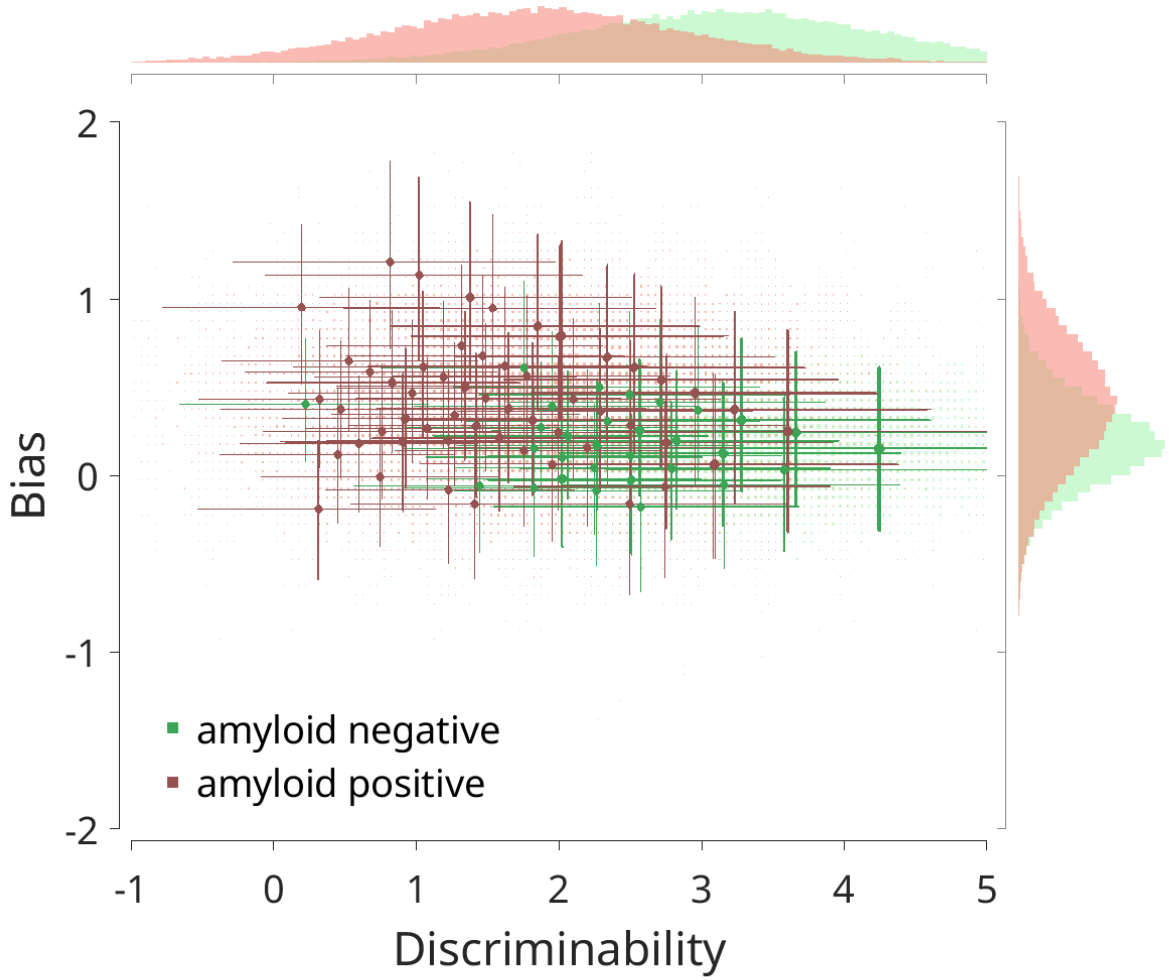
Individual-level Inferences



Signal Detection Group-level Inferences



Signal Detection Individual-level Inferences



References i

Bean, J. (2011). Rey auditory verbal learning test. *Encyclopedia of Clinical Neuropsychology*, 2174–2175.